SUSTAINABLE AGRICULTURE — ENERGY, WATER AND PRODUCTIVITY

DHARA

Distributed Harvest, Agri-Resource and Advisory — from *dhāra*, a flow

An energy-aware control and advisory platform for smallholder irrigation, storage and market access

Architecture and systems-engineering submission

Crop-agnostic, parameterised by agro-climatic zone

Three numbers usually read as three problems

40%

of the national workforce is in agriculture, on holdings that are overwhelmingly under two hectares

90%

of freshwater withdrawals go to irrigation, much of it lifted on subsidised or free electricity

₹1.53L cr

of food is lost after harvest every year, concentrated in fruit, vegetables and oilseeds

They share one cause. At the point of use, neither water nor the energy to lift it carries a price the farmer can see — so nothing in the system rewards using less of either.

Sources: problem statement; ICRIER food-loss estimate for 2020–22; NSO 77th round (89.4% of holdings under 2 ha)

THE FAILURE MODE THIS SYSTEM IS BUILT AGAINST

A solar pump makes energy free at the margin. Unless the design actively prevents it, every efficiency gain is spent pumping more water, not saving any.

Punjab has spent roughly ₹1.25 lakh crore on free farm power since 1997 and about ₹10,000 crore in 2025–26 alone, across some 14 lakh tubewells — while the water table falls close to a metre a year. Efficient hardware alone has never reversed that.

Five design commitments

Commitment	What it forces in the design
Fail safe, never fail shut	No component failure may stop irrigation. Degraded modes water less precisely; they never stop watering.
Offline is the normal case	Every control decision has a local implementation. Losing the cloud costs precision, not function.
Energy is a budget, not a supply	Pumping, cooling and compute are scheduled against a forecast solar harvest, in a fixed priority order.
Water needs a price even when energy has none	Volumetric caps and cluster aquifer feedback stand in for the missing price signal.
Advice must be auditable	Every recommendation carries its inputs, model version and reason code, replayable in the farmer's language.

These are the constraints the rest of the architecture is derived from, not aspirations bolted on afterwards.

Five tiers, each able to run without the one above it

TIER 0 **Field**	Soil probes at 15 / 30 / 60 cm	Line flow meter and pressure	Pump energy meter	Latching solenoid valves
TIER 1 **Farm edge**	Local water-balance loop	Dry-run and stuck-valve interlocks	Thirty-day store and forward	Write-ahead actuation log
TIER 2 **Village cluster**	LoRaWAN gateway, 5 to 10 km	Solar energy arbitration	Cold room and aggregation point	Cached models, runs cloud-free
TIER 3 **Cloud**	Irrigation decision engine	Energy and storage optimiser	Shelf-life and logistics models	Savings verification ledger
TIER 4 **People**	Three-lamp pump box, no reading	Voice call, any feature phone	WhatsApp in the local language	Operator console at the FPO
OUTSIDE **Public systems**	AgriStack farmer and crop registry	IMD forecast, Sentinel-2 imagery	eNAM and Agmarknet prices	Discom, PM-KUSUM, PMKSY

Up: telemetry and verified events. Down: setpoints, advisories, prices. Across: arbitrated solar energy.

TIER 1 — THE PART THAT MATTERS MOST

The farm edge box decides on its own, and proves what it did

What is in the box

- ESP32-class controller, 4 MB flash ring buffer
- LoRa radio, plus an optional 4G module
- Two latching-valve drivers and a pump contactor
- Hall flow input, pressure input, current transformer
- Real-time clock with a week of drift tolerance
- Runs on 2.5 W average from its own 20 W panel

Target bill of materials, installed: about ₹24,500 per farm.

What it decides without asking anyone

- Whether the root zone has crossed its refill threshold
- How many millimetres to apply, and in which window
- Whether the pump is running dry, from flow against current
- Whether a valve failed to seat, from the pressure trace
- When to stop, by measured volume rather than by clock

Every actuation is written to the local log *before* the valve moves, so a reconnect reconciles exactly and the savings ledger is defensible.

A cached seven-day reference-evapotranspiration forecast lets the box keep scheduling properly through a week with no connectivity at all.

The degradation ladder

Each rung is a declared, tested operating mode — not an error path discovered in the field. No rung stops the water.

—	What has failed	How the system behaves	Precision kept
L0	Nothing	Cloud-optimised schedule, aware of forecast and solar	Full
L1	Backhaul to cloud down	Cluster runs the cached model on its stored forecast	About 90%
L2	Cluster gateway down	Edge box runs its own water balance on local sensors	About 75%
L3	Soil probe failed or drifting	Falls back to the crop-calendar schedule, raises a service call	About 50%
L4	Edge box dead	Mechanical bypass; farmer pumps by hand, advice by voice call	Manual, still watered

Precision figures are design targets for the pilot to measure, not results. L4 is the acceptance test: unplug the box and the farm must keep running.

ENERGY ARCHITECTURE

Surplus midday solar is spent in a fixed order, and irrigation is not the last claim on it

1. Crop-critical irrigation, where a stress threshold is crossed
2. Cold-room pull-down for produce already inside
3. Tank fill, capped at the season's volumetric allocation
4. Thermal store charged for the coming night
5. Battery top-up for controls, radio and gateway
6. Export to the feeder under PM-KUSUM, or curtail

Thermal store, not a battery bank

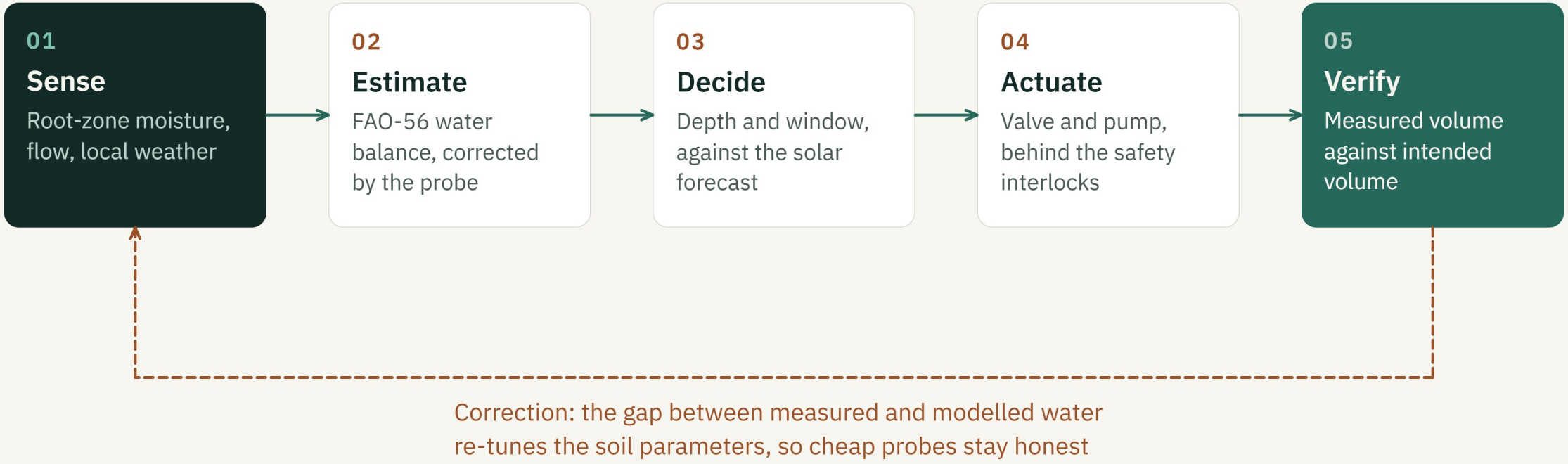
A phase-change ice bank carries the cold room through the night at roughly a third the installed cost of equivalent lithium storage, and it does not lose capacity to cycle count.

Chemistry batteries in this design carry only the controller, radio and gateway — tens of watt-hours, not kilowatt-hours. That is what keeps the cluster bill of materials inside reach of an FPO.

Rule 3 is the one that prevents free energy from becoming wasted water: the tank stops filling at the allocation, whatever the sun is doing.

THE IRRIGATION CONTROL LOOP

Five steps, running every fifteen minutes on the edge box



Step 5 is what makes the savings claim auditable, and what a discom or a carbon buyer is actually paying for.

POST-HARVEST

A shelf-life clock, not a warehouse booking

STEP 01

Pre-cool at the gate

Field heat comes out within hours of harvest in a five-tonne solar cold room shared across the cluster. ICRIER finds each extra kilometre to market adds about 1.4% loss in Bihar; pre-cooling at the farm gate collapses that distance to nothing.

STEP 02

Hold on a clock

Every lot carries its own integrated temperature history, so the platform predicts remaining life per lot rather than per bay. A lot that warmed during a door-open event moves up the dispatch queue automatically.

STEP 03

Dispatch on aggregation

Lots of matching grade pool across farms until a full truck is worth routing. Smallholders get the freight economics of a large grower without having to become one.

11.6% → 7.6%

Modelled loss for a tomato-class crop: a 35% relative reduction, on the assumption that 70% of loss is temperature or handling driven and cold chain removes half of that.

Baseline from the NABCONS 2020–22 study of 54 crops: tomato 8.37% at farm operations plus 3.25% at market level.

FARMER INTERFACE

Four rungs of interface, so literacy and a smartphone are never the entry fee

RUNG 1

Three lamps

On the pump box. Water now, wait, or call for service. No phone, no reading, no network.

RUNG 2

Outbound voice call

Any feature phone. A spoken recommendation with its reason, then a keypad reply. Works on 2G.

RUNG 3

WhatsApp voice notes

Ask a question by speaking it. Answers in the same language, with a photo of the affected leaf if sent.

RUNG 4

Operator console

At the FPO. Cluster water budget, cold-room occupancy, service queue, dispatch planning.

Confidence gating is a hard requirement, not a nicety

An advisory that is wrong once loses the farmer for a season. Below a calibrated confidence threshold the system says it does not know and routes to a human agronomist, rather than guessing fluently. Every answer carries its inputs and model version so a wrong call can be traced and corrected.

DESIGN ARTIFACT — SENSING AND DATA MODEL

Sixteen bytes on the radio, seven entities in the store

Uplink frame, packed binary

Byte	Field	Encoding
0	Frame type, version	uint8
1–2	Sequence number	uint16
3–8	Moisture, 3 depths	uint16, 0.01%
9–10	Soil temperature	int16, 0.1 °C
11–12	Battery	uint16, mV
13–14	Flow counter	uint16, litres
15	Status bits	8 flags

Comfortably inside the 51-byte payload a spreading factor of 12 allows on IN865, so a node still reaches a gateway at the far edge of the cluster on its worst day.

Core entities

Plot — geometry, area, texture, field capacity, wilting point

CropCycle — crop, variety, sowing date, stage, Kc curve

Device — type, firmware, last seen, battery, health state

Reading — device, timestamp, metric, value, quality flag

Decision — input hash, millimetres, reason code, model version

IrrigationEvent — volume, energy, source, decision reference

StorageLot — crop, mass, intake, thermal history, predicted life

Decision and IrrigationEvent are the audit pair: every litre that moved is traceable to the reasoning that asked for it.

DESIGN ARTIFACT — INTERFACE WIREFRAMES

The same recommendation, carried three ways

AT THE BOREWELL

Pump box

- Green — water now
- Amber — wait, soil is wet
- Red — call for service

One button below the lamps: press and the pump runs for the recommended volume, then stops itself.

6:40 AM, OUTBOUND CALL

Voice, any handset

"Namaskar. Your tomato plot needs 18 millimetres today, best between 10 and 12."

"Soil at 30 centimetres has dropped to 41 percent. No rain expected for four days."

"Press 1 to start the pump. Press 2 to hear this again. Press 9 for a person."

Reason before instruction, every time. The farmer is the controller, not the controlled.

WHATSAPP THREAD

Ask by speaking

Voice note, 7 seconds — "these leaves are curling, what is it?"

"Send me one photo of the underside of a curled leaf."

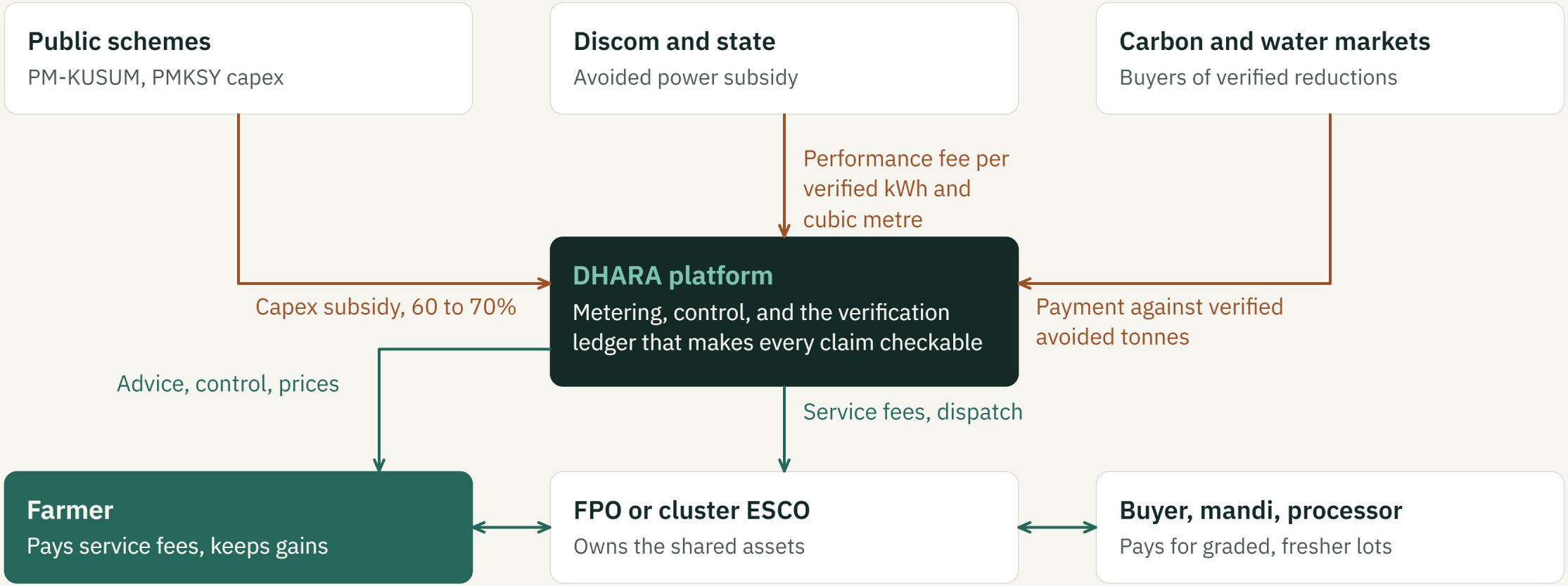
"I am not certain from this photo. I am passing it to Sunita at the FPO, she will call before noon."

The third bubble is the important one. Saying so beats guessing fluently.

Wireframes are schematic. Language, voice and call timing are per-cluster configuration, set with the FPO at onboarding.

DESIGN ARTIFACT — MONEY FLOW

The farmer is not asked to pay for a saving that lands on someone else's books



Terracotta: money in, earned by measurement. Teal: services and produce moving between the people on the ground.

THE STATED BASELINE

Every saving on the next slide is measured against exactly this

Parameter	Baseline value	Where it comes from
Holding size	1.2 ha	NSO 77th round
Seasonal crop evapotranspiration	450 mm	Reference vegetable
Field application efficiency	0.45, flood or furrow	Standard range 0.40 to 0.50
Over-application from blind scheduling	plus 20%	Assumption, tested opposite
Pumping head	40 m	Assumption
Pumpset wire-to-water efficiency	0.35	Typical unrated Indian set
Marketable yield	25 t/ha	Reference vegetable
Farm-gate price	₹12 per kg	Assumption
Post-harvest loss	11.62%	NABCONS 2020–22, tomato
Grid emission factor	0.710 tCO2 per MWh	CEA baseline v21.0, FY25

Crop, soil and climate are configuration. Swapping a crop swaps evapotranspiration, the Kc curve, yield, price and the loss profile; the architecture does not change.

QUANTIFIED BENEFIT, PER HECTARE PER SEASON

Full stack: efficient conveyance, sensed scheduling, solar lift, cold chain

6,631
cubic metres of water
12,000 down to 5,369. A 55% cut in applied water.

2,354
kilowatt-hours
3,737 down to 1,383. A 63% cut in lift energy.

1.67 t
CO2 equivalent
2.65 t if the remaining lift is solar rather than grid.

1,017
kilograms kept
Loss 11.6% to 7.6%, worth about ₹12,200.

One cluster, 100 farms, 120 hectares, one season

0.80 million cubic metres of water · 283 MWh · 200 tonnes CO2e · 122 tonnes of produce that reaches a buyer instead of a ditch

The 55% figure sits inside the 51 to 66% pumping-hour reductions the PMKSY Per Drop More Crop evaluation measured in Pune, Saharanpur and Nalgonda.

ATTRIBUTION

What the software earns, separated from what the pipe earns

5,864 m3

The drip retrofit. 48.9 of the 55.3 percentage points. Hardware anyone can buy, financed under PMKSY.

767 m3

Sensing and scheduling. 6.4 points on top — 12.5% of what a drip farm still over-applies.

If drip farms over-apply by	Software alone saves
10%	256 m3/ha — 4.5%
15%	511 m3/ha — 8.7%
20%, the case used here	767 m3/ha — 12.5%
25%	1,023 m3/ha — 16.0%
30%	1,278 m3/ha — 19.2%

Why the smaller number is still the one that matters

Drip has been subsidised for two decades and adoption has not, on its own, bent the groundwater curve. What the platform adds is the part that was missing: a measured, per-plot record of how much water and energy actually moved, and why.

That record is what lets a discom pay for a negawatt, a carbon buyer pay for a tonne, and a regulator enforce a cap. The 6.4 points are the saving. The measurement is the product.

RISK REGISTER

The first row is the one most proposals leave out

Risk	Why it bites	What the design does about it
Rebound on saved water	Freed water is spent expanding irrigated area, so the aquifer keeps falling	Volumetric allocation per plot, cluster aquifer feedback, caps written into the service contract
Free power kills the incentive	Whoever saves the energy is not whoever pays for it	The discom pays the performance fee; the farmer's return is produce, not kilowatt-hours
Cheap sensors drift	A confidently wrong moisture reading is worse than no reading	Self-calibration against measured irrigation events, cross-node checks, a quality flag on every value
Field hardware walks away	Nodes sit unattended in open fields for months	FPO ownership, deliberately low resale value, tamper bit in the status byte
One wrong advisory	A single bad call loses a farmer for the whole season	Confidence gating, escalation to a human agronomist, full input and model-version audit trail

Efficiency without a cap has repeatedly increased total withdrawal. Treating that as the headline risk rather than a footnote is a design decision.

TARGET FARMER PROFILES

Not every farmer gets hardware, and saying so is part of the design

PROFILE A

Irrigated horticulture

0.8 to 2 ha, own borewell, drip already on part of the holding, growing vegetables or fruit for a mandi within a day's reach.

Full stack. Highest value per hectare, real post-harvest losses to recover, fastest payback. This is where the pilot goes.

PROFILE B

Irrigated field crops

1 to 2 ha of cereals or cane on free or heavily subsidised power, flood irrigated, pumping whenever supply is available.

Full stack, but gated. The largest absolute water and subsidy savings, and no farmer-side business case at all until a discom contract exists.

PROFILE C

Rainfed and marginal

Under 1 ha, no assured irrigation, cotton, pulses or millets, most exposed to a late or broken monsoon.

Advisory only. No field hardware, because the payback is not there. Sowing-window, weather and price advice carried by the cluster's existing gateway.

Profile C is the largest group and the least monetisable. It is served because the cluster infrastructure is already paid for by A and B, not because a separate business case was invented for it.

DEPLOYMENT AND SCALE-UP

Four phases, each with a gate that can fail

Phase	Window	Scale	Gate to the next phase
0 · Instrument	Months 0–6	One cluster, 100 farms, with paired control plots	A measured baseline exists, not a modelled one
1 · Prove	Months 6–18	10 clusters, 1,000 farms, 3 districts, 2 states	Claimed savings survive an independent audit
2 · Contract	Months 18–36	100 clusters, 15,000 farms	A discom signs a performance contract on verified savings
3 · Scale	Months 36–60	1,000 clusters, 150,000 farms	Registry integration at state level, not per-farm onboarding

Phase 0 is the one that is usually skipped

Without paired control plots and a season of pre-intervention metering, every later savings claim is a model output arguing with itself. The Digital Agriculture Mission's farmer registry and digital crop survey give the cohort definition; the edge ledger gives the measurement. Nothing is claimed before both exist.

UNIT ECONOMICS

Sold as a service, because no smallholder should buy this outright

Platform capital cost	Rupees
On each farm	
Soil node, three depths, LoRa	6,000
Edge controller and drivers	8,000
Valves, flow and pressure sensing	6,500
Install and commissioning	4,000
Shared across 100 farms	
LoRaWAN gateway, solar, mast	85,000
Weather micro-station	45,000
Five-tonne solar cold room	13,50,000
Operator console at the FPO	25,000
Platform capex per farm, 1.2 ha	39,550

What the farmer pays

Irrigation as a service, ₹1,200 per hectare per season. Cold storage at ₹2.50 per kilogram for the first week, then ₹0.30 per kilogram per day. Advisory free at the point of use.

What the farmer gets back

About ₹14,600 a season on a 1.2 hectare holding from avoided post-harvest loss alone, before any price-timing gain. Simple payback on platform capex: **2.7 seasons**.

What the state gets back

₹16,500 per hectare per season of avoided supply cost at ₹7 per kilowatt-hour, rising to ₹26,200 where the residual lift goes solar. For scale: Punjab currently carries roughly ₹71,000 per tubewell per year.

The solar pump and the drip system are financed separately under PM-KUSUM and PMKSY and are not carried in this table.

WHERE THIS STANDS

The architecture is the easy part. The measurement is what makes anyone pay for it.

Grounded in evidence

Loss percentages from the NABCONS study of 54 crops. Water and energy savings triangulated against the PMKSY Per Drop More Crop evaluation. Emission factor from the CEA baseline database. Subsidy figures from Punjab's own budget.

Still assumed

Pumping head, farm-gate price, the 20% over-application factor and the share of post-harvest loss that cold chain can actually reach. Every one is a slider in the model, and Phase 0 exists to replace them with measurements.

What would be built first

The edge controller and its degradation ladder, then the verification ledger. Not the dashboard, not the model. If the box cannot be trusted alone in a field for a season with no network, nothing above it matters.

DHARA — a working prototype of the edge control loop and the benefit model is the natural next step.